

Review

Spinning disc reactors: recent advances in CFD modeling for optimization and scale-up

Jatupon Yencham¹, Anh N Phan², Weerinda Mens³,
 Sompong Putivisutisak⁴ and Suttichai Assabumrungrat^{1,5}



A spinning disc reactor (SDR) is an intensified chemical reactor renowned for its excellent heat and mass transfer performance. Its applications have expanded to a wide range of chemical processes, including nanoparticle synthesis and both chemical and photochemical reactions. Computational fluid dynamics (CFD) has become an essential tool for understanding SDR hydrodynamics and guiding reactor design, optimization, and scale-up. This review critically examines the development of models for predicting liquid film thickness and the key CFD modeling approaches, including multiphase flow, turbulence modeling, and mixing characterization. It also discusses current challenges in SDR scale-up and proposes future research directions to support the broader industrial implementation of this promising SDR technology.

Addresses

¹ Center of Excellence in Catalysis and Catalytic Reaction Engineering, Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University, Bangkok 10330, Thailand

² School of Engineering, Newcastle University, Newcastle Upon Tyne NE1 7RU, United Kingdom

³ Department of Chemical and Materials Engineering, Faculty of Engineering, Rajamangala University of Technology Thanyaburi, Pathumtani 12110, Thailand

⁴ Advanced Computational Fluid Dynamics Research Unit, Department of Mechanical Engineering, Faculty of Engineering, Chulalongkorn University, Bangkok 10330, Thailand

⁵ Bio-Circular-Green-economy Technology & Engineering Center (BCGeTEC), Faculty of Engineering, Chulalongkorn University, Bangkok 10330 Thailand

Corresponding author: Assabumrungrat, Suttichai
 (Suttichai.A@chula.ac.th)

Current Opinion in Chemical Engineering 2026, 52:101261

This review comes from a themed issue on **Anchoring computational flow models to real-world reactors**

Edited by **Vivek Ranade** and **Ashwin Patwardhan**

For complete overview of the section, please refer to the article collection, "[Anchoring computational flow models to real-world reactors](#)"

Available online 8 May 2026

<https://doi.org/10.1016/j.coche.2026.101261>

2211–3398/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Introduction

In recent years, spinning disc reactors (SDRs), intensified chemical reactors using a rapidly rotating disc to create a thin and highly sheared liquid film, have been applied in a wide range of chemical processes, such as nanoparticle precipitation [1–4], polymerization [5,6], biodiesel production [7–9], glycerolysis [10], enzymatic reactions [11], and liquid-liquid (L-L) extraction [12]. Compared with conventional reactors (e.g. stirred tank reactors), SDRs offer advantages such as compact size, enhanced safety, cleaner operation, and improved efficiency [13,14]. These benefits arise from the formation of a thin liquid film over the rotating disc, where strong centrifugal forces generate high shear rates, significantly enhancing heat and mass transfer while enabling very short residence times [10,15,16]. Liquid film thickness is therefore a key parameter governing heat and mass transfer, mixing efficiency, and overall reactor performance. A brief summary of models for predicting liquid film thickness and its measurement was provided in [Appendix A2](#) and [Table A2](#).

From a reactor-engineering perspective, the significance of SDRs lies not only in their thin-film hydrodynamics but also in how the flow field governs the hierarchy of mixing scales. In general, mixing in chemical reactors occurs sequentially across three mixing length scales: macromixing (bulk flow distribution), mesomixing (turbulent dispersion), and micromixing, where concentration gradients are eliminated at the molecular scale [1,2]. In SDRs, thin liquid films, high shear and centrifugal forces enhance micromixing, which is important for fast competitive and series-parallel reactions, where local mixing may affect product selectivity rather than overall conversion [3,4]. The characteristic timescale of this process is the micromixing time (t_m), and reported SDR values are generally much shorter than those of mechanically stirred reactors, with a comparative summary provided in [Appendix A2](#) ([Table A2](#)). Accordingly, the benefit of SDRs should be interpreted not only in terms of intensified transport but also in terms of the relative magnitude of the reaction, residence and micromixing timescales.

From a practical standpoint, SDRs occupy an intermediate position between stirred reactors and highly miniaturized continuous-flow devices. They offer

Nomenclature		Acronyms	
<i>Symbols</i>		E_k	Ekman number.
D	Disc radius (cm).	Re	Reynolds number.
d_{SMD}	Sauter mean diameter of droplets (mm).	Gr	Grashof number.
Z	Gap size between top and bottom surfaces (cm).	Sc	Schmidt number.
h_{avg}	Mean liquid film thickness (cm).	Pe	Peclet number.
$Vol.$	Volume fraction (-).	PI	Process intensification.
v	Velocity (m/s).	SDR	Spinning disc reactor.
v_r	Radial velocity (m/s).	CFD	Computational fluid dynamics.
t_{res}	Mean residence time (s).	DNS	Direct numerical simulation.
t_m	Micromixing time (s).	LES	Large eddy simulation.
k	Turbulence kinetic energy (J/kg).	RANs	Reynolds-averaged Navier-Stokes.
ε	Turbulence dissipation rate (m^2/s^3).	CSF	Continuum surface force.
ω	Specific dissipation rate (1/s).	CFL	Courant-Friedrichs-Lewy or Courant number.
k_{La}	The overall mass transfer coefficient (1/s).	VOF	Volume of fluid.
n	Tank-in-series number.	SST	Shear Stress Transport.
σ_0^2	Dimensionless variance.	SF	Smooth film.
<i>Greek letters</i>		CSW	Concentric wave.
ν	Kinematic viscosity (Pa.s).	SW	Spiral wave.
λ	Radial length scale factor.	RTD	Residence time distribution.
ρ	Liquid density (kg/m^3).	PINNs	Physics-informed neural networks.
μ	Dynamic viscosity (m^2/s).	SM	Smooth disc.
ω	Energy dissipation rate (W/kg).	ES	Elliptic-cylindrical spoiler discs.
σ	Surface tension (N/m).	CS	Cylindrical spoiler discs.
		TEA	Techno-economic analysis.
		DOE	Design of experiments.

stronger transport intensification than stirred tanks, but often greater flexibility and throughput than micro-channel systems [5,6]. However, most reported SDR applications remain at laboratory scale, and broader industrial deployment is still limited by factors such as disc size, production rate, hydrodynamic stability, and the challenge of maintaining favorable liquid-film behavior during scale-up and numbering-up [5,7]. Representative throughput ranges and pilot-scale examples are summarized in [Appendix A3 \(Table A3\)](#).

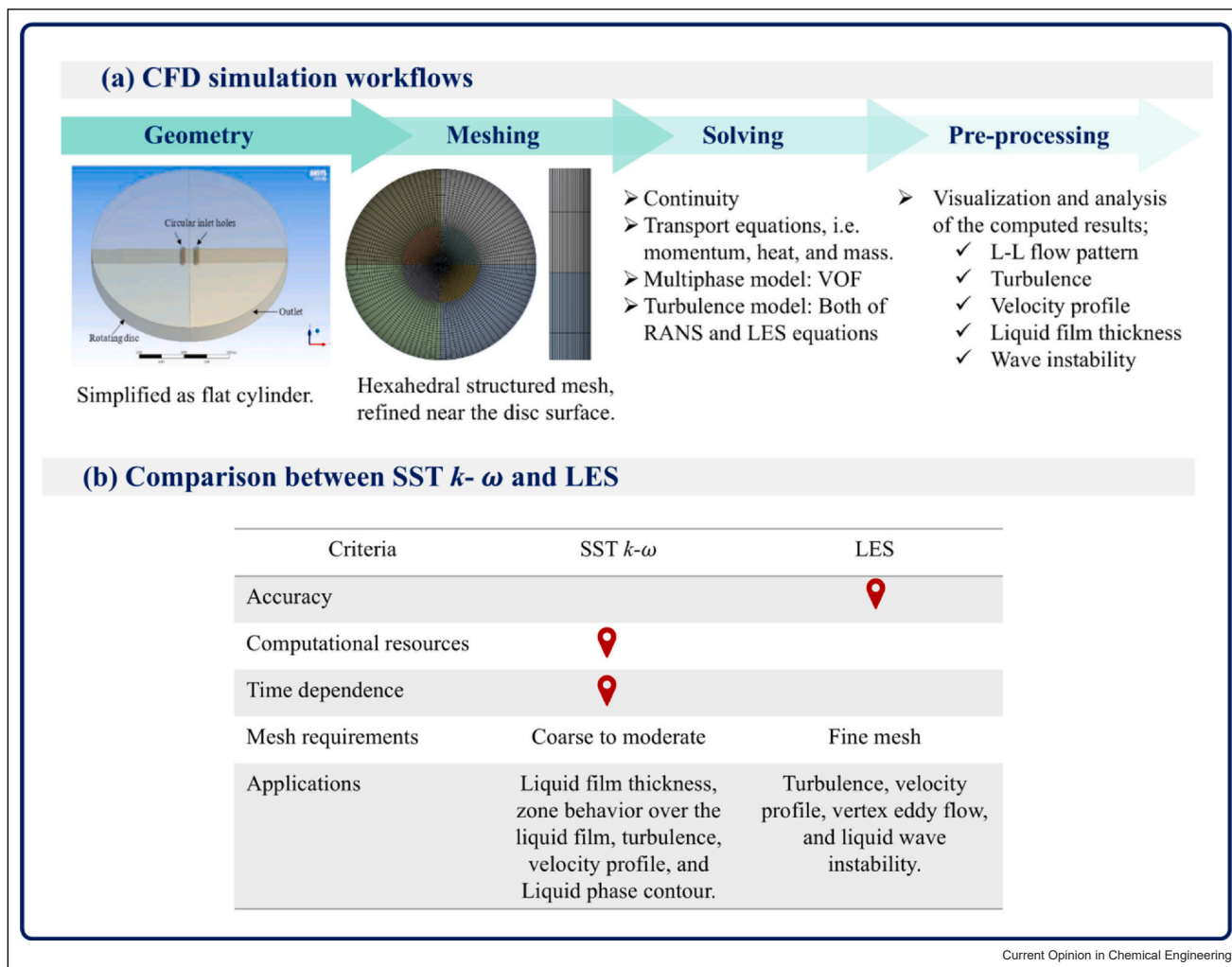
These practical constraints make predictive design tools increasingly important for SDR development. Computational fluid dynamics (CFD) has become a valuable approach for analyzing liquid-film morphology, interfacial instabilities, wettability, residence-time behavior, and other hydrodynamic behaviors that are difficult and costly to characterize experimentally over a wide operating range [8–13]. CFD has played a key role in intensified reactor development, providing practical insights for process optimization and scale-up, including in SDR systems. This review discusses hydrodynamic constraints relevant to scale-up, reviews CFD strategies for SDR design and optimization, and highlights decision-relevant outputs that can support industrial implementation.

Role of CFD in SDR design, optimization, and scale-up

Before discussing specific CFD strategies, it is important to recognize the hydrodynamic constraints that govern SDR performance. At the macroscopic level, the liquid film is commonly described in terms of injection, acceleration, and synchronization zones, with the synchronization zone often being the most extensive region and closely associated with mixing enhancement and energy dissipation [11]. However, the intensified conditions that improve micromixing may also shorten residence time or destabilize the film. These trade-offs are commonly interpreted using the Reynolds number (Re), Weber number (We), and Ekman number (E_k), together with the relevant reaction and residence timescales. Hence, successful scale-up requires predictive tools that can identify operating windows compatible with both hydrodynamic stability and reactor performance. Further details of hydrodynamic zones, flow regimes, and dimensionless-group interpretation are provided in [Appendix A4](#) and [Figures A1 and A2](#).

CFD has therefore become an important tool for analyzing thin-film hydrodynamics and transport processes in SDRs [9–16]. In SDR studies, the value of CFD lies

Figure 1



Summary of SDR-CFD simulation setups; **(a)** CFD simulation workflows and **(b)** Comparison between SST $k-\omega$ and LES models.

not only in visualizing complex flow behavior but also in predicting decision-relevant quantities such as film thickness, wetted area, wave development, residence-time characteristics, and turbulence-related descriptors that can be linked to reactor performance. This section outlines CFD modeling strategies for SDRs in ANSYS Fluent, including key numerical setup considerations and relevant outputs such as film thickness, shear stress, and residence time characteristics. The typical SDR-CFD workflow is illustrated in Figure 1.

Geometrical domain simplification and meshing

For CFD analysis, the SDR is commonly simplified to a flat cylindrical domain focusing on liquid-film flow. Gas-phase effects, including aerodynamic drag, are often neglected due to the relatively large gap between the rotating disc and the housing, resulting in weak velocity

gradients near the upper boundary [12,17–20]. Nevertheless, this assumption should be used with caution for enclosed or industrial-scale systems, where omission of the housing or stator may underrepresent gas-phase shear effects. A circular inlet hole is created on the top surface at the disc center, with a diameter matching the inlet pipe, and defined as a velocity inlet. The remaining top surface and side boundaries are defined as pressure outlets, while the bottom surface is treated as a rotating wall. Mesh types commonly used include structured, unstructured, and hybrid meshes. Most SDR models are meshed with a hexahedral structured mesh because they provide uniform grid distribution and high numerical accuracy for simple geometries, with lower computational cost [31]. However, SDR models with surface modifications were set with a tetrahedral unstructured mesh, owing to its greater flexibility for complex geometries, but it may introduce numerical errors and

higher computational effort [32]. Mesh refinement near the disc surface is essential to resolve velocity gradients and eddy structures within the liquid film. In addition, mesh requirements depend on operating conditions. Higher rotational speeds and lower viscosities (thinner films) require finer interfacial resolution, while lower speeds or higher viscosities allow coarser meshes. For modified disc surfaces, the mesh conforms to the surface geometry. To balance accuracy with computational cost, grid independence analyses are conducted, typically using liquid film thickness or average parameters such as mean film thickness or Sauter mean diameter (SMD), with a deviation below 5% considered acceptable [12,28,33].

Multiphase models

Interface modeling is an essential component of SDR-CFD simulations, as liquid flow over the spinning disc involves air–liquid multiphase interactions. The volume of fluid (VOF) model is therefore widely used to capture the liquid–air interface in SDR simulations [9–13,15,18]. Surface tension effects are incorporated into the momentum equation to accurately represent liquid-film wave dynamics, typically using the continuum surface force model to track the air–liquid and liquid–liquid interfaces and accurately account for body forces at the interface and the wall [10–13].

Generally, the VOF can be discretized using either an explicit scheme [10,12,13] or an implicit scheme [9,15,18]. The explicit scheme is most used to discretize the VOF equations for the SDR-CFD. It generally requires lower computational cost per iteration but requires a small step size, whereas the implicit scheme allows for larger time steps at the expense of higher computational cost [21]. The time step size is typically set between 2.5×10^{-5} s and 1.0×10^{-3} s for the CFD-SDR model under the laboratory scale. To be practical at the large-scale operation of SDR processes, the adaptive time step by controlling the Courant–Friedrichs–Lewy or Courant number (CFL), defined as the ratio of the distance a fluid travels in one time step to the mesh grid size, is considered to ensure numerical stability and reduce the overdesign time step size to minimize the simulation time [22]. Although specific limits for SDRs are not well established, values below ~ 0.15 have been reported for rotating systems [23]. Therefore, a maximum CFL of ~ 0.2 is recommended for stable transient simulations, but it should be verified by time-step sensitivity analysis for the specific SDR case.

Viscous models

The choice between Reynolds-Averaged Navier–Stokes (RANS) and Large Eddy Simulation (LES) in SDR-CFD depends on the required level of hydrodynamic detail and computational cost. For most engineering and

scale-up applications, RANS models are preferred due to their lower computational demand and their ability to predict key time-averaged hydrodynamic features, such as liquid film thickness, velocity profiles, residence time, and overall flow-zone behavior. In contrast, LES is recommended when detailed resolution of transient flow structures and interfacial instabilities is required, particularly for studies involving micromixing and small-scale wave dynamics, albeit at significantly higher computational cost.

As summarized in Table 1, SDR-CFD studies mainly employ RANS and LES approaches. RANS model uses time-averaged equations with turbulence closure based on eddy viscosity (μ_t), typically requiring additional transport equations for turbulence kinetic energy (k) and either the turbulence dissipation rate (ϵ) or the specific dissipation rate (ω). However, because turbulent fluctuations are modeled rather than resolved, RANS is less suitable for resolving unsteady large-scale eddies or detailed interfacial turbulence structures. In contrast, LES applies filtered Navier–Stokes equations to resolve large turbulent structures, with subgrid-scale models representing smaller eddies. LES is therefore more suitable when the objective depends on interfacial instabilities, wave-induced eddies, or micromixing-sensitive flow features, albeit at substantially higher computational cost [9,10,12,13,15].

Among RANS approaches, the SST k – ω model is widely applied in SDR-CFD studies, due to its ability to combine the near-wall treatment of the k – ω model with the robustness of the k – ϵ model in the free-stream region through a blending function [9,10,12]. It is commonly used for predicting liquid film thickness and flow-zone behavior, typically with meshes of approximately 500,000 elements for simplified domains (Table 1). In addition, some studies have investigated small-scale eddies and wave instabilities within the liquid film across different flow zones using LES. Such analyses typically require very fine meshes (~ 10 million elements), resulting in significantly higher computational cost [13]. Therefore, the SST k – ω model is generally preferred for practical SDR-CFD simulations, while LES is reserved for detailed analyses of flow instabilities.

Common sources of error in SDR-CFD simulations

Although CFD can reduce experimental burden and support cost-effective design exploration [24,25], SDR-CFD predictions still contain important sources of uncertainty that must be considered to ensure reliable and reproducible results. In general, these uncertainties arise from numerical setup, interface-capturing strategy, turbulence closure, mesh resolution, and wetting-related

Table 1

Summary of geometrical parameters of SDR and its meshing.

Year	Geometrical parameter	Meshing	Grid independent parameter ^b	Model	Applications	Ref.
2012	D = 85 mm Z = 5 mm	- Structured mesh - 540,000 elements	h_{avg} ^a	VOF	Examines liquid film thickness, turbulence through specific dissipation rate, and species transport to characterize mixing in the liquid film.	[15]
2015	D = 54 mm Z = 5 mm	- Unstructured mesh - 756,000 elements	d_{SMD}	VOF	Investigate the droplet size of molten slag breakup off the spinning disc edge	[12]
2019	D = 200 mm Z = 2 mm	- Structured mesh - 14,000,000 elements	Wave instability	SST $k-\omega$	Investigates liquid film instabilities over the spinning disc, including concentric, spiral sine-like, and spiral pulse-like waves, and provides correlations for transitions between these regimes.	[13]
2020	D = 54 mm Z = 5 mm	- Structured mesh - A grid size of 40 $\mu m \times 10 \mu m$	V_r	VOF laminar	Examines liquid film thickness and radial velocity under varying disc speed, viscosity, feed-pipe diameter, and flow rate. Presents a synchronization-radius correlation validated by high-speed camera measurements.	[11]
2022	D = 130 mm Z = 3 mm	- Unstructured mesh - 469,550 elements	h_{avg} , Vol_{Hept} , ω .	VOF SST $k-\omega$	Examines liquid film thickness, zone behavior, and turbulence (via energy dissipation rate) to clarify the mechanisms behind improved SDR extraction efficiency.	[9]
2023	D = 200 mm Z = 150 mm.	- Structured mesh - 15,000,000 elements	h -profile	VOF SST $k-\omega$	Examines liquid film thickness, radial velocity, and turbulence, using average and radial profiles of turbulent kinetic energy to elucidate mixing efficiency improvements.	[16]

^a Computational domain was discretized into seventeen cylindrical sections with heights ranging from 5 μm to 150 μm , varying from the inlet to the disc edge.

^b Selected variables for grid independence test.

input parameters. A major source of uncertainty in SDR simulations is the treatment of the liquid–gas interface. Because the VOF model is widely used to track the air–liquid interface in SDR-CFD studies [9–13,15,18], prediction accuracy depends strongly on grid resolution, interface reconstruction, and time-step size. Both explicit and implicit schemes have been used in previous studies [10,12,13]. Explicit schemes can preserve a sharper interface but require smaller CFL-controlled time steps, whereas implicit schemes allow larger time steps at the cost of greater numerical diffusion [21]. In practice, explicit schemes are often used with first-order discretization, while implicit schemes can adopt second-order formulations for improved temporal accuracy. Therefore, careful selection of numerical settings is required to balance accuracy and computational cost.

Another commonly reported discrepancy is the under-prediction of liquid film thickness compared to experimental measurements (e.g. laser-based techniques) [10,11,14,19]. This is mainly related to turbulence modeling. The SST $k-\omega$ model, while effective in capturing near-wall behavior and overall flow fields, is based on RANS formulations and therefore cannot resolve small-scale eddies and interfacial instabilities [10,13]. As a result, surface waviness and film curvature may be underestimated, leading to deviations from experimental observations. Although higher-fidelity approaches such as LES or Direct Numerical Simulation can provide improved resolution of turbulent structures and interfacial dynamics, their computational cost remains prohibitive for practical SDR applications, particularly in large-scale or parametric studies [13]. Therefore, the SST $k-\omega$ model represents a practical compromise, providing reasonably accurate predictions of velocity fields and turbulence quantities (e.g. turbulent kinetic energy and dissipation rate) at an acceptable computational cost.

In addition, grid resolution is also closely related to CFL control and convergence behavior. Mesh resolution and convergence behavior introduce further uncertainty. Finer meshes generally require smaller time steps to maintain stability and reduce numerical diffusion, but excessive refinement may lead to slow convergence or residual stagnation and substantially increase computational cost. Grid-independence testing is therefore essential. For film-thickness validation, CFD predictions are commonly compared with theoretical correlations such as Nusselt-based models, as discussed in Appendix A2. Furthermore, the wettability of the liquid film over the rotating disc remains a challenge in CFD modeling. While CFD modeling can provide reasonable predictions of liquid film thickness, it may not accurately capture liquid film breakage under conditions of very high rotational disc speed and low volumetric flow rate. Previous studies have examined flow transitions in

spinning disk reactors using a laser velocimeter with an uncertainty of $\pm 0.04\%$. However, a general correlation for predicting film breakage has not yet been established. Therefore, the critical condition for film breakage should be further studied through experiments and numerical analysis to develop a wettability correlation based on dimensionless numbers such as Weber (We), Reynolds (Re), and Ekman (E_k).

In summary, CFD–SDR modeling involves uncertainties from numerical schemes, turbulence modeling, grid resolution, and input parameters. Although reasonable predictions can be obtained, limitations remain in capturing interfacial behavior and film breakage. Therefore, careful numerical setup and validation are essential to ensure reliable results.

Optimization and key CFD outputs

CFD simulation is a powerful tool for optimizing the PI technologies, including rotating packed reactor [24,26], microchannel reactor [27], fluidized bed reactor [28], and SDR [9–13,15,18]. Its key feature lies in enabling detailed analysis and optimization of reactor geometry and flow hydrodynamics within the complex, small-volume configurations characteristic of PI systems [29]. By resolving key transport phenomena, CFD supports improved reactor design and operating conditions. A schematic of process optimization parameters is shown in Figure 2. When properly applied, it reduces development time, accelerates scale-up, and enhances overall process economics [24,26]. For SDR processes, CFD-based optimization generally addresses three linked questions: (i) how operating conditions can be adjusted to maintain a thin and stable liquid film [11,14,18], (ii) how rotor-disc geometry can be modified to intensify mixing and mass transfer [9,10,30], and (iii) how residence time is distributed over the spinning disc [16]. In this sense, CFD is valuable not only for describing thin-film flow but also for identifying operating windows and design modifications that improve reactor performance under transport-limited conditions. An example of hydrodynamic optimization is provided by water denitrification studies, in which transient CFD simulations were used to relate film thickness, energy dissipation rate, micromixing time, and residence time to nitrate-removal efficiency [18]. Under the reported optimum conditions, the SDR achieved 85% nitrate removal at 250 mL/min and 1403 rpm, corresponding to a τ_m of 0.001 s, an energy dissipation rate of 122 W/kg, and a residence time of 0.24 s. This example highlights how CFD can identify intensified operating conditions that combine short residence time with high process performance.

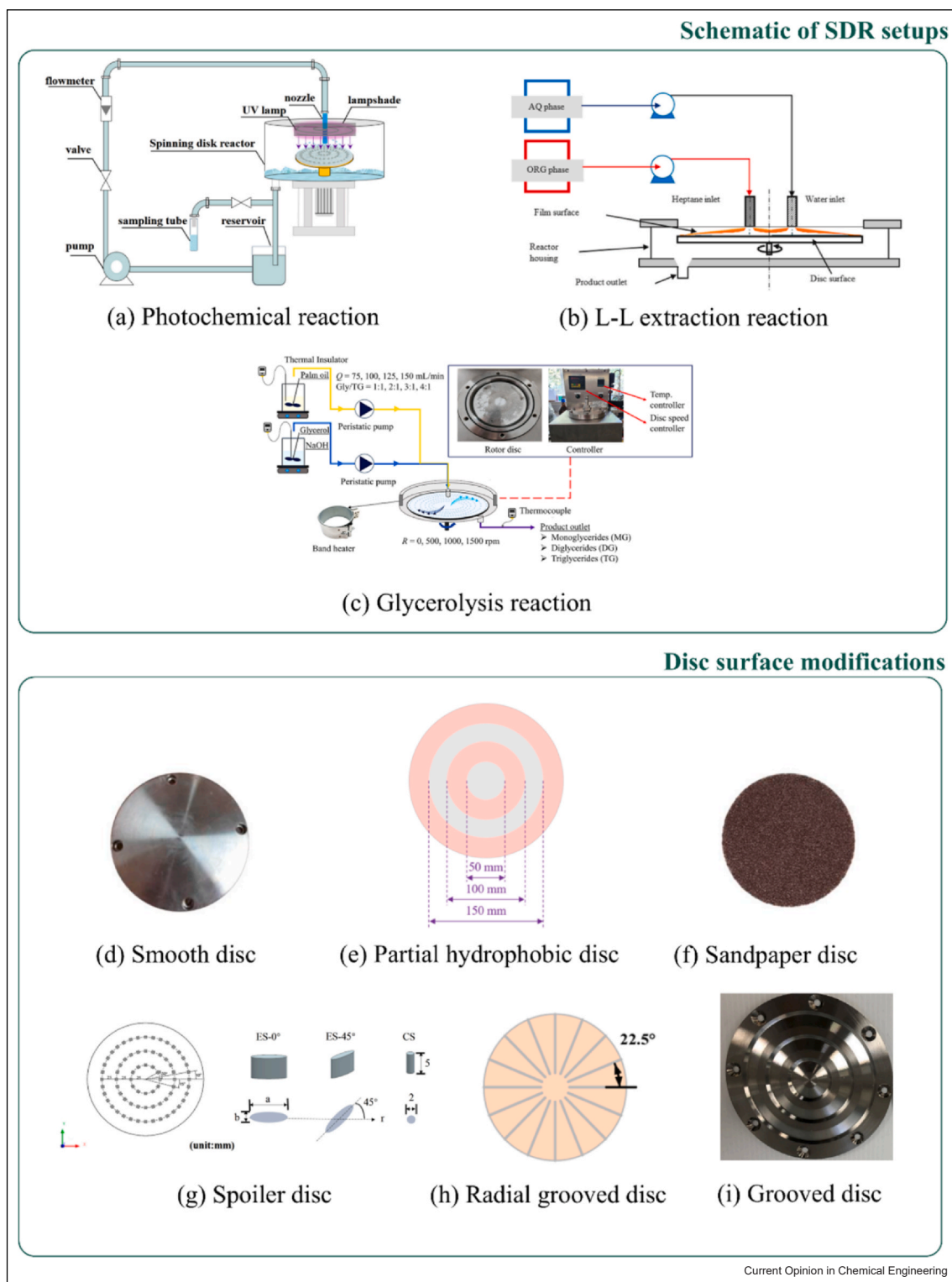
CFD has also been applied to geometry-driven optimization. Various disc modifications, such as smooth partial

hydrophobic disc [12,27], sandpaper disc [12], spoiler disc [28], radial grooved disc [29], and grooved disc [30], have been used to enhance mixing (Figure 2d–i). These modifications aim to intensify micromixing within the thin liquid film, improving the process's yield. Accordingly, the turbulence parameters (e.g., turbulent kinetic energy and energy dissipation rate) are considered together with performance indicators such as k_La and reaction yield, in comparison with smooth discs. Surface roughness modifications, such as sandpaper coatings, have achieved extraction efficiencies of up to 70% and k_La values around 9.6 s^{-1} , associated with thinner films and enhanced energy dissipation rate [9]. These studies show that CFD can be used not only to visualize flow patterns, but also to explain why specific surface modifications improve reactor performance.

Beyond local hydrodynamic and turbulence descriptors, residence-time-distribution (RTD) analysis provides an important reactor-level interpretation of CFD results. In SDRs, RTD determines how long reactants experience the intensified thin-film environment and how closely the reactor approaches plug-flow or dispersed-flow behavior. These characteristics are directly relevant to conversion, selectivity, and scale-up, particularly when disc modifications alter recirculation and flow non-ideality. For RTD-oriented SDR studies, a pulse-tracer method has been used to compare smooth discs (SM), elliptic-cylindrical spoiler discs (ES), and cylindrical spoiler discs (CS) [16]. Water-like fluids were introduced over the rotating disc. The simulations were performed with the numerical setup of the VOF, SST k – ϵ , and species transport. A transient flow field was first solved without species transport until steady state was achieved, after which a tracer pulse was injected. The mean residence time (t_r), dimensionless variance (σ_θ^2), tank-in-series number (n), and Peclet number (Pe) were determined, and correlations with We and Re were proposed. The result showed that structured discs (ES and CS) exhibited higher t_r (0.29–1.43 s) and variance (0.02–0.15) than the SM, indicating greater dispersion. The ES configuration showed the greatest recirculation and the largest deviation from ideal plug-flow behavior, followed by the CS, whereas the SM approached plug-flow behavior more closely. Therefore, these RTD-derived quantities provide useful reduced-order descriptors for interpreting disc design, reactor nonideality, and scale-up.

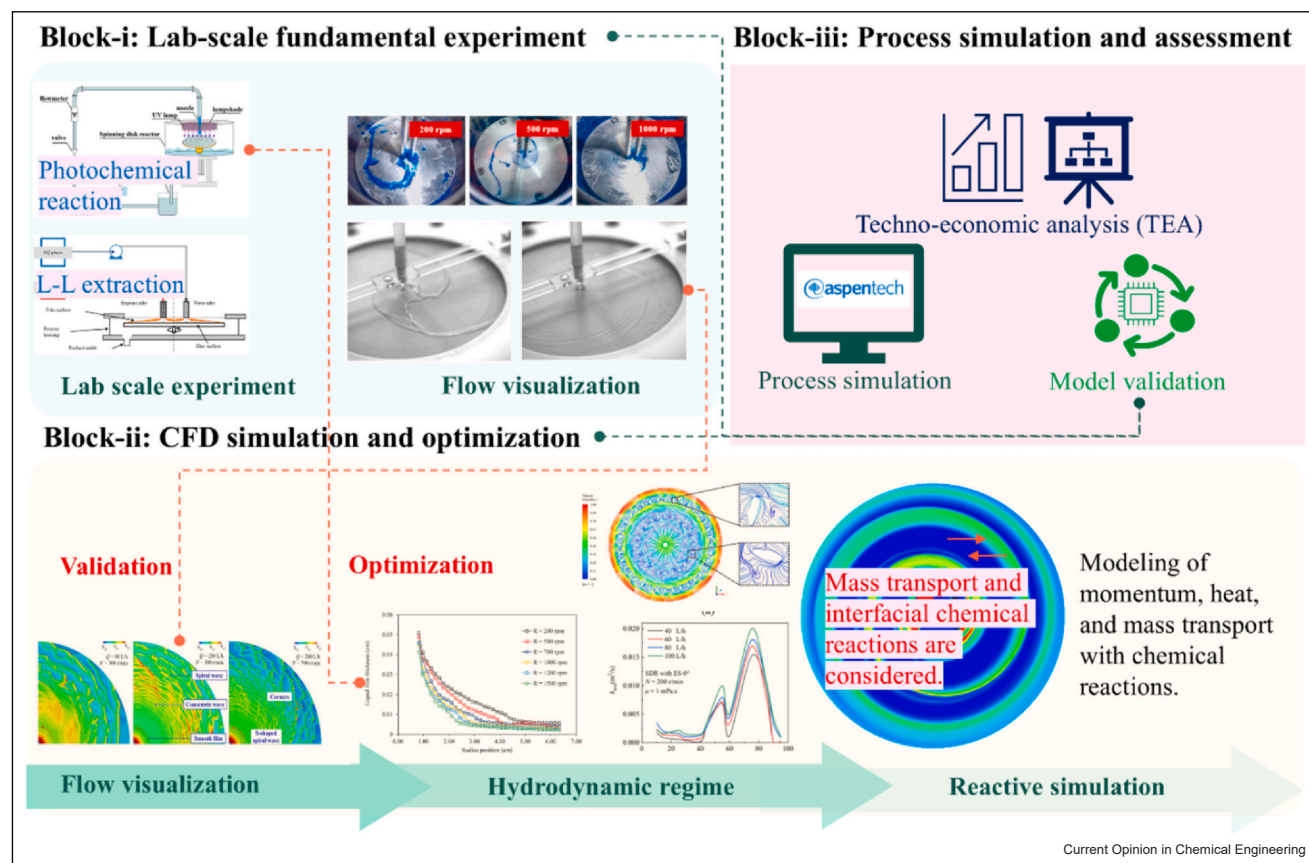
Overall, current SDR studies show that CFD-derived hydrodynamic descriptors can be linked to process performance. However, generalized frameworks combining these descriptors with dimensionless-group analysis remain limited. Future work should integrate key CFD outputs with Reynolds, Weber, Ekman, and Schmidt numbers to support more transferable design and scale-up.

Figure 2



Schematic of SDR setups and disc surface modifications: (a-c) Applications in photochemical reaction [10], Liquid-liquid extraction [9], and Glycerolysis reaction [31]; (d-i) Smooth disc [9], Partial hydrophobic disc [30], Sandpaper disc [9], Spoiler disc [10], Radial grooved disc [32], and Grooved disc [33].

Figure 3



Three-block methodology for developing process-intensified (PI) equipment [26].

Future perspectives and recommendations

SDRs offer strong potential for transport-intensified applications, but broader implementation remains limited by scale-up and production-capacity constraints [3,34]. Although production rates of about 300 kg/h have been reported for 1 m discs, this remains insufficient for bulk chemical applications [6,35]. Consequently, the feasibility of SDR scale-up must be carefully evaluated, as discussed in the previous section. To advance SDRs toward industrial implementation, the three-stage process intensification (PI) development framework shown in Figure 3 should be considered [26]. A three-stage PI development framework was proposed in 2022 [26], comprising (i) lab-scale fundamental experiments, (ii) CFD simulations & optimization, and (iii) process simulations and assessment. Stage (i) evaluates reactor performance through experiments, flow visualization, and design of experiments (DOE). Stage (ii) uses CFD to investigate hydrodynamics and reactive behavior, optimize operating conditions and reactor design, validate results with experimental data, and assess scale-up in a cost-effective manner. Stage (iii) applies process simulation tools (e.g. Aspen Plus, Pro-II) together with

CFD-derived scale-up data to perform techno-economic analysis (TEA).

SDR technology has largely completed stage (i) for several green chemistry applications, including biodiesel production [36], polymerization [35,37], photochemical reactions [8,10,38], and glycerolysis [31], demonstrating high yields and reliable laboratory-scale experimental validation. Current development is primarily focused on stage (ii), where combined experimental and CFD are used to optimize disc surface modifications [8–10], as mentioned previously section. In particular, maintaining favorable liquid-film hydrodynamics, especially synchronization-zone behavior, residence time, and wettability, remains critical for SDR performance [6,9–11,31,38]. These phenomena directly influence transport and reaction performance but remain insufficiently characterized. However, CFD studies of SDRs that explicitly incorporate reactive systems and mass transfer modeling are still limited. This lack of validated data and correlations restricts the ability to translate SDR performance into process simulation tools (e.g., Aspen Plus), where reliable inputs such as effective

mass transfer coefficients, apparent kinetics, and residence time distribution (RTD) are required. As a result, scale-up and broader industrial implementation remain slow.

In contrast, several process intensification technologies have successfully progressed toward industrial implementation. Both process scale-up and numbering-up were applied for the industrial adoption. For example, the methanol-to-olefins (MTO) process in fluidized bed reactors has been scaled from laboratory to commercial scale using CFD-based approaches [39]. Simulations across multiple reactor sizes, from micro-scale to commercial units, revealed significant changes in hydrodynamics during scale-up. While small-scale reactors showed uniform feed distribution, larger systems required gas distributors to maintain proper flow uniformity. Additionally, flow regimes evolved from homogeneous structures to bubbling and particle clustering at larger scales, leading to altered product distributions, including increased coke formation and reduced ethylene yield. These findings show that hydrodynamics and reaction performance are closely linked, highlighting the important role of reactive CFD in maintaining lab-scale performance at an industrial scale. For the numbering-up, most numbering-up studies have focused on microchannel reactors, where the key concerns are flow uniformity and pressure-drop control [40]. Among the two main strategies, external numbering-up appears more practical for SDRs because it is simpler and more scalable, although flow maldistribution remains an important challenge. It should be noted that SDR scale-up would likely require both disc enlargement and numbering-up, together with careful control of hydrodynamic similarity and feed distribution across units.

Future work should prioritize high-fidelity CFD-SDR simulations incorporating reactive and mass species transport to generate physically consistent datasets that can be reduced into low-order (0D/1D) models for key design parameters, such as mass transfer coefficients (k_{La}) and reaction kinetics. Building on prior studies of rotating packed bed absorbers, where detailed transport phenomena were translated into process simulation (e.g., Aspen Plus) via rate-based models, the governing correlations were formulated based on penetration theory and expressed through relevant dimensionless groups, including the interfacial area ratio, Schmidt number (Sc), Reynolds number (Re), and Grashof number (Gr) [41]. These correlations were implemented through Intel® Visual Fortran to capture mass transfer, interfacial area, and liquid holdup [41]. For SDR systems, these correlations, particularly those governing mass transfer, interfacial area, and liquid holdup, should be systematically linked to liquid film thickness and residence time, and extended to additional dimensionless

groups (e.g., Weber and Ekman numbers) [13,14,31]. A hybrid DOE–CFD framework enhanced with physics-informed neural networks is recommended to accelerate design and operational exploration, enabling scale-up via disc enlargement or numbering-up, followed by TEA to assess industrial feasibility.

CRedit authorship contribution statement

Jatupon Yencham: Conceptualization, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Anh N Phan:** Conceptualization, Formal analysis, Writing – review & editing. **Weerinda Mens:** Conceptualization, Formal analysis, Visualization, Writing – original draft, Writing – review & editing, Project administration. **Sompong Putvisutisak:** Conceptualization, Formal analysis, Writing – review & editing. **Suttichai Assabumrungrat:** Conceptualization, Writing – review & editing, Funding acquisition, Resources, Supervision.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to express appreciation for the support of the NRCT-JSPS Grant via the National Research Council of Thailand (NRCT) [Grant number N11A670925], and the Ratchadaphiseksomphot Endowment Fund, Chulalongkorn University (The Review Article Grant No.RA68003). Additionally, we are grateful to Mr. Panut Bumphenkiattikul of SCG Chemicals Co. Ltd. for his excellent advice on CFD modeling.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.coche.2026.101261](https://doi.org/10.1016/j.coche.2026.101261).

References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- of special interest
 - of outstanding interest
1. Yao L-B, Wu W, Wu X-S, Chu G-W, Luo Y, Sun B-C: **Intensification of micromixing efficiency in a spinning disk reactor: experimental investigation.** *Chem Eng Process Process Intensif* 2021, **166**:108500.
 2. von Damnitz L, Anders D: **A review on the mixing quality of static mixers.** *ChemEngineering* 2025, **9**:128.
 3. Boodhoo KVK, Al-Hengari SR: **Micromixing characteristics in a small-scale spinning disk reactor.** *Chem Eng Technol* 2012, **35**:1229-1237.

4. van Kouwen ER, Winkenwerder W, Brentzel Z, Joyce B, Pagano T, Jovic S, Bargeman G, van der Schaaf J: **The mixing sensitivity of toluene and ethylbenzene sulfonation using fuming sulfuric acid studied in a rotor-stator spinning disc reactor.** *Chem Eng Process Process Intensif* 2021, **160**:108303.
5. Boodhoo K: **Spinning disc reactor for green processing and synthesis.** *Process Intensification for Green Chemistry.* John Wiley & Sons; 2013:59-90.
6. Mens W: **Spinning disc reactor: status, challenges and future perspectives for advancement.** *Chem Eng Process Process Intensif* 2024, **205**:109977.
7. Oxley P, Brechtelsbauer C, Ricard F, Lewis N, Ramshaw C: **Evaluation of spinning disc reactor technology for the manufacture of pharmaceuticals.** *Ind Eng Chem Res* 2000, **39**:2175-2182.
8. Qiu Z, Petera J, Weatherley LR: **Biodiesel synthesis in an intensified spinning disc reactor.** *Chem Eng J* 2012, **210**:597-609.
9. Mens-Appamama W, Yenham J, Putivisutisak S, Bumphenkiattikul P, Vongachariya A, Khaoodee W, Ngaosuwan K, Wongsawaeng D, Assabumrungrat S: **Investigation of mass transfer and hydrodynamics of liquid-liquid extraction in spinning disc reactor by computational fluid dynamics simulation.** *Results Eng* 2022, **16**:100798.
- This work investigated mass transfer and hydrodynamics in liquid-liquid extraction within an SDR using CFD simulation, thereby providing valuable insight into the interplay between flow behavior and extraction performance.
10. Li Z-H, Li Y-B, Zhu Y-G, Xu Y-C, Sun B-C, Zou H-K, Chu G-W: **Regulating liquid film disturbance via disk configurations for photochemical process intensification in a spinning disc reactor.** *Chem Eng Sci* 2023, **273**:118641.
- This work demonstrates that using an SDR for photochemical reactions enhances photon absorption and light penetration through intense thin-film mixing. Disc surface spoilers further improve turbulence and reaction performance, as confirmed by coupled CFD simulations and experiments.
11. Wang D, Jin H, Ling X, Peng H, Yu J, Cui Z: **Regulation of velocity zoning behaviour and hydraulic jump of impinging jet flow on a spinning disc reactor.** *Chem Eng J* 2020, **390**:124392.
12. Wang D, Ling X, Peng H: **Simulation of ligament mode breakup of molten slag by spinning disk in the dry granulation process.** *Appl Therm Eng* 2015, **84**:437-447.
13. Li Y, Wu XS, Yazhao L, Chu G-W, Sun BC, Luo Y, Chen J: **Three-dimensional large eddy simulation of wave characteristics of liquid film flow in a spinning disc reactor.** *AIChE J* 2020, **66**:e16894.
14. Wang D, Gu C, Sun C, Ling X, Peng H, Yang X, Yuan F, Du J, Yu W: **Flow pattern transition and wave characteristics of impinging jet flow on a spinning disc reactor.** *Chem Eng J* 2023, **451**:138790.
15. de Caprariis B, Di Rita M, Stoller M, Verdone N, Chianese A: **Reaction-precipitation by a spinning disc reactor: influence of hydrodynamics on nanoparticles production.** *Chem Eng Sci* 2012, **76**:73-80.
16. Li Z-H, Xu H-Z, Li Y-B, Sun B-C, Chu G-W: **Determination of liquid residence time characteristics in spinning disc reactors via a computational approach.** *Ind Eng Chem Res* 2023, **62**:15635-15647.
- This work uses CFD to accurately predict RTD in SDRs and resolve experimental measurement discrepancies. Disc spoilers significantly increase mean residence time and flow non-ideality compared to a flat disc. RTD correlations were developed with deviations within $\pm 20\%$, linking hydrodynamics to operating conditions.
17. Julia Simone DK: **Reaction Engineering and Hydrodynamics of a Rotor-stator Spinning Disc Reactor.** Technical University of Munich; 2019:10.
18. Vilardi G, De Caprariis B, Stoller M, Di Palma L, Verdone N: **Intensified water denitrification by means of a spinning disc reactor and stirred tank in series: kinetic modelling and computational fluid dynamics.** *J Water Process Eng* 2020, **34**:101147.
19. Vilardi G, Stoller M, Di Palma L, Boodhoo K, Verdone N: **Metallic iron nanoparticles intensified production by spinning disk reactor: optimization and fluid dynamics modelling.** *Chem Eng Process Process Intensif* 2019, **146**:107683.
20. Sana S, Manson J, Zivkovic V, Boodhoo K: **Solvent-antisolvent precipitation in a spinning disc reactor: precipitation kinetics and mathematical modelling of starch nanoparticles.** *Chem Eng Process Process Intensif* 2022, **176**:108962.
21. ANSYS Fluent Theory Guide, Release 2021 R2. ANSYS, Inc; 2021.
22. Ferziger JH, Perić M, Street RL: *Computational Methods for Fluid Dynamics.* Fourth Edition, Springer; 2020.
23. Trivellato F, Castelli MR: **On the Courant-Friedrichs-Lewy criterion of rotating grids in 2D vertical-axis wind turbine analysis.** *Renew Energy* 2014, **62**:53-62.
24. Yang Y-C, Ouyang Y, Zhang N, Yu Q-J, Arowo M: **A review on computational fluid dynamic simulation for rotating packed beds.** *J Chem Technol Biotechnol* 2019, **94**:1017-1031.
25. Lodh B: **The transformative role of Computational Fluid Dynamics (CFD) in chemical engineering.** *Open J Chem* 2024, **10**:001-003.
- This work highlights the transformative role of CFD in chemical engineering, emphasizing its value as a tool for reactor analysis, process optimization, and design improvement across a wide range of applications.
26. Ouyang Y, Heynderickx GJ, Van Geem KM: **Development of intensified reactors: a process intensification methodology perspective.** *Chem Eng Process Process Intensif* 2022, **181**:109164.
27. Bravo L, Gutierrez CR, Cobo M, Figueredo M: **CFD evaluation of a microchannel reactor for ammonia synthesis using a Ruthenium-based catalyst.** *Chem Eng Process Process Intensif* 2026, **219**:110624.
28. Yu X, Blanco PH, Makkawi Y, Bridgwater AV: **CFD and experimental studies on a circulating fluidised bed reactor for biomass gasification.** *Chem Eng Process Process Intensif* 2018, **130**:284-295.
29. Ramírez-Márquez C, Al-Thubaiti MM, Martín M, El-Halwagi MM, Ponce-Ortega JM: **Processes intensification for sustainability: prospects and opportunities.** *Ind Eng Chem Res* 2023, **62**:2428-2443.
30. Yao L-B, Liu W, Wang X-T, Li Z-H, Wu X-S, Sun B-C, Chu G-W: **Feasibility study on micromixing intensification in a spinning disc reactor utilizing heterogeneous surface wettability.** *Chem Eng Process Process Intensif* 2022, **170**:108707.
31. Yenham J, Putivisutisak S, Weeranoppanant N, Ngaosuwan K, Chanthan N, Zullaikah S, Phan AN, Assabumrungrat S, Mens W: **Intensified glycerolysis of palm oil in a spinning disc reactor for continuous monoglyceride production.** *ACS Sustain Chem Eng* 2025, **13**:11301-11314.
32. Wang D, Qiu Y, Yuan F, Chen Z, Yang X, Du J, Yu W, Ling X, Peng H: **Film flow and micromixing behaviours of a turbulent jet impinging on a radially grooved spinning disc reactor.** *Chem Eng Process Process Intensif* 2025, **213**:110314.
- This work investigates enhanced mixing in an SDR using radially grooved discs under high volumetric flow rates. The Villiermaux-Dushman reaction is used to quantify mixing performance, showing that increased groove number and angle promote wavelet instabilities. These instabilities enable effective mixing at lower rotational speeds, improving productivity while reducing energy consumption.
33. Carr T, Russo Abegão F, Boodhoo K: **Purification of hemicellulose hydrolysates by antisolvent precipitation in a spinning disc reactor, Biofuels.** *Bioprod Bioref* 2024, **18**:952-967.
34. Jacobsen NC, Hinrichsen O: **Micromixing efficiency of a spinning disk reactor.** *Ind Eng Chem Res* 2012, **51**:11643-11652.
35. Boodhoo K, Jachuck R: **Process intensification: spinning disc reactor for condensation polymerisation.** *Green Chem* 2000, **2**:235-244.

36. Appamana W, Sukjareem P, Ngaosuwan K, Assabumrungrat S: **Intensification of continuous biodiesel production using a spinning disc reactor.** *J Chem Eng Jpn* 2019, **52**:545-553.
37. Pask SD, Nuyken O, Cai Z: **The spinning disk reactor: an example of a process intensification technology for polymers and particles.** *Polym Chem* 2012, **3**:2698-2707.
38. Li Z-H, Li Y-B, Sun B-C, Chu G-W, Chen J-F: **Modeling and experimental verification for photodegradation in a spinning disk reactor.** *Chem Eng J* 2024, **493**:152634.
39. Lu B, Zhang J, Luo H, Wang W, Li H, Ye M, Liu Z, Li J: **Numerical simulation of scale-up effects of methanol-to-olefins fluidized bed reactors.** *Chem Eng Sci* 2017, **171**:244-255.
40. Dong Z, Wen Z, Zhao F, Kuhn S, Noël T: **Scale-up of micro- and milli-reactors: an overview of strategies, design principles and applications.** *Chem Eng Sci X* 2021, **10**:100097.
41. Joel AS, Wang M, Ramshaw C: **Modelling and simulation of intensified absorber for post-combustion CO₂ capture using different mass transfer correlations.** *Appl Therm Eng* 2015, **74**:47-53.